Brownian Motion:
From Theory to Proof of Molecular Reality

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1. INTRODUCTION

During the late nineteenth century, theoretical physics grappled with fundamental unresolved issues. While debates over electromagnetism and relativity dominated, Robert Brown's 1827 observation of the irregular motion of pollen grains suspended in water—now termed Brownian motion—quietly posed a challenge to prevailing thought [1]. At that time, the atomic hypothesis underpinning Ludwig Boltzmann's statistical thermodynamics faced significant skepticism. Boltzmann's framework, which derived thermodynamic laws from the ensemble dynamics of material atoms, had proponents but also notable critics, including Max Planck and Wilhelm Ostwald, who questioned the existence of atoms [2, 3]. Some physicists, such as Wilhelm Roentgen, interpreted Brownian motion as a potential inconsistency in Boltzmann's second law, suggesting it contradicted thermodynamic equilibrium.

Albert Einstein's 1905 paper offered a transformative explanation. He proposed that the random motion of suspended particles resulted from thermal collisions with molecules of the surrounding liquid [4]. These molecular impacts drove the motion while viscosity provided damping, leading to a quantifiable mean square displacement, $\langle x^2 \rangle = 2Dt$, where D is the diffusion coefficient. Beyond affirming the reality of atoms, Einstein's formulation enabled the determination of Avogadro's number, linking thermal energy to observable mechanical displacement. This quantitative connection to statistical mechanics effectively resolved doubts about Boltzmann's theory [4].

Concurrently, Marian Smoluchowski's 1906 statistical model and Paul Langevin's 1908 dynamical approach reinforced Einstein's findings [5, 6]. Jean Perrin's 1908 experiments provided definitive validation, measuring particle displacements to confirm the theoretical predictions and earning him the 1926 Nobel Prize [7]. Einstein's work, indirectly recognized in his 1921 Nobel Prize for the photoelectric effect, solidified the atomic theory [8]. Together, these efforts established Brownian motion as a probe of molecular kinetics.

Today, Brownian motion is recognized as a key driver of microscale biological processes, such as nutrient diffusion and cellular signaling. This report unveils the unfolding story of Brownian motion: from the foundational insights of Einstein and Smoluchowski, which set the stage, to the strikingly clear and dynamic framework of Langevin, whose approach takes center stage, culminating in Perrin's meticulous experiments that brought the theory to life.

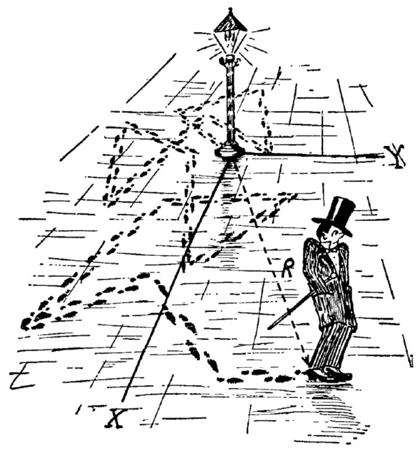

FIG. 1. A whimsical depiction of a drunkard's random walk under a streetlamp—an early and intuitive analogy often used to explain Brownian motion. The dashed paths represent the unpredictable steps taken, illustrating the concept of a random walk in two dimensions. (*Image Credit*: Computation Unit, Advanced Design Studies, University of Tokyo).

2. THEORY

Imagine a tiny particle, like a grain of pollen, suspended in water and constantly jostled by the random collisions of water molecules. Suppose the particle starts at a fixed position, say a point we'll call the origin. As time passes, the particle takes a series of random steps in all directions due to these molecular collisions, much like a drunkard stumbling in random directions due to impaired coordination from excessive drinking. After certain time, we ask: on average, how far away from its initial position has the particle gone? Albert Einstein (1905) first derived the mean square displacement of a Brownian particle using a diffusion framework, while Paul Langevin (1908) introduced a dynamical approach with a stochastic force. Langevin described his approach to Brownian motion as "infinitely more simple" than Einstein's. Given its apparent simplicity — and since Einstein's treatment has already been covered in class — I will focus on Langevin's derivation of the mean squared displacement, drawing additional clarity from *The Feynman Lectures on Physics* [9].

2.1. Derivation of the Random Walk

We approach to Brownian motion proceeds in two stages: first, a random walk model to establish the diffusive nature of the motion, and second, a dynamical equation to derive the mean square displacement quantitatively. I begin with the common intuitive random walk model, which captures the essence of Brownian motion as a series of random displacements. Consider a particle undergoing a random walk in one dimensions, taking n steps, each of length L. At each step, the particle moves in a random direction, so the displacement for the i-th step is a vector $\mathbf{s_i}$ with magnitude $|\mathbf{s_i}| = L$. The directions are uniformly distributed both left and right in 1D space, making the walk isotropic. The position after n steps is the vector sum of the displacements:

$$\mathbf{R_n} = \sum_{i=1}^{n} \mathbf{s_i}$$

The mean square displacement is the expectation value of the squared distance from the origin, which is the inner product of $\vec{R}_n$ with itself:

$$\langle (\mathbf{R_n})^2 \rangle = \langle \mathbf{R_n}, \mathbf{R_n} \rangle = \left\langle \left(\sum_{i=1}^{n} \mathbf{s_i} \right), \left(\sum_{j=1}^{n} \mathbf{s_j} \right) \right\rangle = \sum_{i=1}^{n} \sum_{j=1}^{n} \langle \mathbf{s_i}, \mathbf{s_j} \rangle$$

Since the steps are independent and the directions are random, the expectation of the inner product $\langle \mathbf{s_i}, \mathbf{s_j} \rangle$ depends on whether $i = j$. For $i \neq j$, the directions of $\mathbf{s_i}$ and $\mathbf{s_j}$ are uncorrelated, and the average over all possible directions in 1D gives $\langle \mathbf{s_i}, \mathbf{s_j} \rangle = 0$. For $i = j$, the inner product is the square of the step length: $\langle \mathbf{s_i}, \mathbf{s_j} \rangle = L^2$. Thus:

$$\langle (\mathbf{R_n})^2 \rangle = \sum_{i=1}^{n} \sum_{j=1}^{n} \langle \mathbf{s_i}, \mathbf{s_j} \rangle = \sum_{i=1}^{n} \langle \mathbf{s_i}, \mathbf{s_i} \rangle = \sum_{i=1}^{n} L^2 = nL^2 \tag{1}$$

This result holds in one dimensions. In three dimensions, the motion in each direction is independent, so the total mean square displacement is:

$$\langle (\mathbf{R_n})^2 \rangle = \langle \mathbf{x}^2 \rangle + \langle \mathbf{y}^2 \rangle + \langle \mathbf{z}^2 \rangle = 3nL^2$$

The number of steps n is proportional to time t, as more time allows for more collisions with surrounding molecules. This suggests:

$$\langle (\mathbf{R_n})^2 \rangle = \beta\, t \tag{2}$$

This linear dependence on time is a hallmark of diffusive processes, which I will now quantify using Langevin's dynamical approach.

2.2. Dynamical Equation

To simplify the analysis, let us consider motion in just one dimension and compute the mean of x^2 as a preparatory step (since the motion is symmetric in all directions, the mean values of x^2, y^2, and z^2 are equal. Thus, the mean square of the total displacement in three dimensions is simply three times the result we obtain for one dimension). To compute the coefficient β in Eq. 2, an external force must be introduced, temporarily setting aside the stochastic nature of Brownian motion. This external force elicits a response from the particle governed by two primary effects. First, the particle experiences inertia, characterized by an effective mass m, which may differ from its actual mass due to the displacement of the surrounding fluid. In one-dimensional motion, this inertial effect contributes a term $m\frac{d^2x}{dt^2}$. Second, a steady external force on the particle induces a drag force from the surrounding fluid, which is proportional to the particle's velocity. This arises due to the fluid's viscosity and flow dynamics, and is given by Stokes' formula $6\pi\eta a\frac{dx}{dt}$, where η is the dynamic viscosity of the fluid and a is the radius of the particle. This drag introduces dissipative losses, which are crucial for thermal fluctuations - without dissipation, the kT term associated with thermal energy would not emerge, as fluctuations and dissipation are intrinsically linked via the fluctuation-dissipation theorem [10]. Then, the equation of motion under an external force is

$$m\frac{d^2x}{dt^2} + 6\pi\eta a\frac{dx}{dt} = F_{\text{ext}}, \tag{3}$$

To find the mean square displacement, we multiply both sides of the equation by x and take the average:

$$\left\langle mx\frac{d^2x}{dt^2}\right\rangle + \left\langle 6\pi\eta ax\frac{dx}{dt}\right\rangle = \langle xF_{\text{ext}}\rangle \tag{4}$$

On the right-hand side, the random force is uncorrelated with the position over long times, resulting in

$$\langle xF_{\text{ext}}\rangle = 0. \tag{5}$$

For the second term, involving the drag force, the computation yields

$$\left\langle x\frac{dx}{dt}\right\rangle = \frac{1}{2}\frac{d}{dt}\langle x^2\rangle, \tag{6}$$

since

$$\frac{d}{dt}(x^2) = 2x\frac{dx}{dt} \quad \Rightarrow \quad \left\langle \frac{d}{dt}(x^2) \right\rangle = \frac{d}{dt}\langle x^2 \rangle = 2\left\langle x\frac{dx}{dt} \right\rangle.$$

Finally, for the first term, which is the inertial term, the expression becomes

$$x\frac{d^2x}{dt^2} = x\frac{dv}{dt} = \frac{d}{dt}(xv) - v\frac{dx}{dt} = \frac{d}{dt}(xv) - v^2 \quad \Rightarrow \quad \left\langle x\frac{d^2x}{dt^2} \right\rangle = \frac{d}{dt}\langle xv \rangle - \langle v^2 \rangle$$

Over long times, the term $\langle xv \rangle$ (the correlation between position and velocity) oscillates but does not grow systematically, so its time derivative averages to zero (i.e., $\frac{d}{dt}\langle xv \rangle = 0$). Thus:

$$\left\langle x\frac{d^2x}{dt^2} \right\rangle = -\langle v^2 \rangle. \tag{7}$$

Substituting Eqs. 5, 6, and 7 in Eq. 4, we get

$$-m\langle v^2 \rangle + 3\pi\eta a\frac{d}{dt}\langle x^2 \rangle = 0$$

Using the equipartition theorem, we relate the mean square velocity to the temperature. For a particle in one-dimensional thermal equilibrium, the theorem gives $\frac{1}{2}m\langle v^2 \rangle = \frac{1}{2}k_BT$, where k_B is Boltzmann's constant and T is the absolute temperature. This implies

$$\frac{d\langle x^2 \rangle}{dt} = \frac{k_BT}{3\pi\eta a}$$

Integrating with the initial condition $\langle x^2 \rangle = 0$ at $t = 0$, we find that the mean square displacement after a time t is given by

$$\langle x^2 \rangle = \frac{k_BT}{3\pi\eta a}t \tag{8}$$

This result gives the mean square displacement in one dimension. In three dimensions, since the motion in each direction is independent:

$$\langle R^2 \rangle = \frac{k_BT}{\pi\mu a}t \tag{9}$$

To interpret this result in terms of a diffusion process, we compare Eq. 9 with the standard form for mean square displacement in 3D:

$$\langle R^2 \rangle = 6Dt, \tag{10}$$

where D is the diffusion coefficient. By matching the expressions, we identify

$$D = \frac{k_B T}{6\pi\eta a},$$

which is the celebrated Stokes–Einstein–Sutherland formula [11]. It expresses the diffusion coefficient D in terms of measurable physical parameters: Boltzmann's constant k_B, temperature T, fluid viscosity η, and the particle radius a.

This confirms the linear dependence on time suggested by the random walk model. Thus, we can indeed quantify the distance these particles travel! To apply this in practice, one must first determine how the particle responds to a steady force—specifically, how fast it drifts under a known force—in order to find η. Once that is known, the extent of the particle's random motion can be quantitatively predicted.

Historically, this equation played a crucial role because it provided one of the first experimental pathways to determine Boltzmann's constant k_B. Since parameters like η, time, and displacement can be measured, it became possible to compute k through statistical averaging. This was significant because, in the ideal gas law $PV = RT$, the gas constant R is related to Boltzmann's constant via $R = N_A k_B$, where N_A is Avogadro's number. At the time, R could be measured, but the number of atoms in a mole, N_A, was not precisely known. Thus, careful observation of microscopic particles undergoing Brownian motion allowed scientists to estimate both k_B and N_A, shedding light on the size and quantity of atoms—questions that were once purely philosophical.

3. EXPERIMENTAL OBSERVATION

Smoluchowski [5] proposed a more direct method than Einstein's two successive derivations to calculate the mean square displacement. His approach yielded an expression for $\langle R^2 \rangle$ similar in form to Eq. 9, but with a coefficient larger by a factor of 64/27. Langevin later identified an error in Smoluchowski's execution, despite the soundness of his method, and corrected the calculation, demonstrating that it also leads to Eq. 9. An early experimental attempt to verify these theoretical predictions was conducted by Svedberg [12]. His measurements of $\langle R^2 \rangle$ deviated from Eq. 9 by a factor of approximately 4, aligning more closely with Smoluchowski's uncorrected result. Langevin noted that Svedberg's measurements were not direct and that the Brownian particles observed were likely too small for Stokes' law to apply, a key assumption underlying Eq. 9. Despite this discrepancy, Langevin placed greater trust in the well-founded theories of Einstein and himself, supported by rigorous derivations, over the apparent agreement between Smoluchowski's flawed calculation and Svedberg's imperfect experiment.

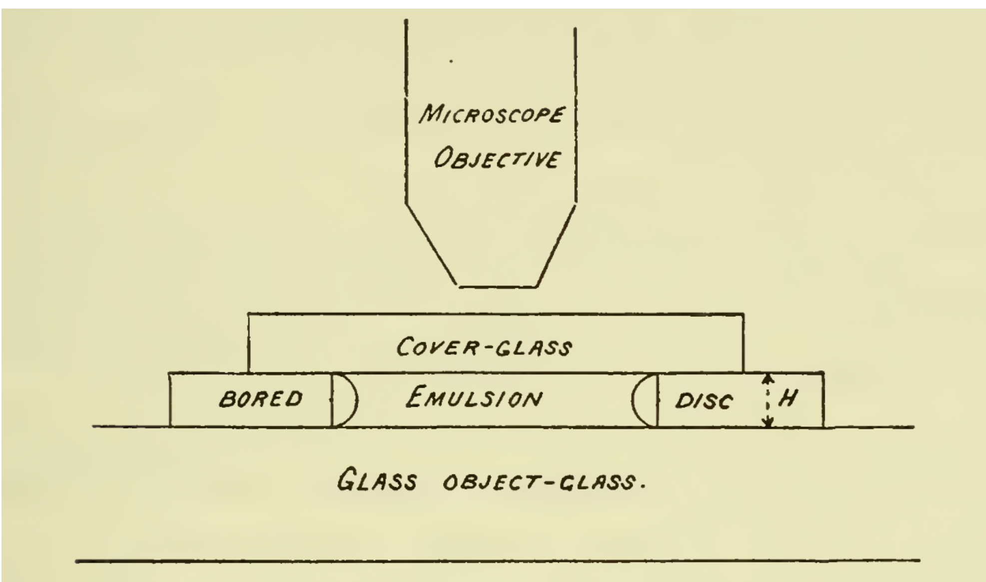

FIG. 2. Perrin's experimental setup for observing Brownian motion. A thin glass plate with a central hole was fixed onto a glass slide to form a sealed chamber. A drop of the colloidal emulsion was placed at the center and flattened with a cover glass, which adhered to the upper surface of the perforated plate, thereby closing the cell. To prevent evaporation, the edges of the cover glass were sealed with paraffin or varnish, allowing the preparation to remain stable for several days or even weeks. The sealed chamber was then placed on the stage of a carefully leveled microscope. By adjusting the focus vertically, Perrin was able to observe granules at different depths within the fluid [7].

Langevin's clarification of the theory not only resolved earlier inconsistencies but also helped inspire Jean Perrin to undertake a more rigorous experimental investigation [14]. In 1908, Perrin transformed theoretical predictions into empirical evidence through a series of painstaking experiments [7]. His setup was simple: a suspension, a microscope, and patience (see Fig. 2). Using emulsions of gamboge and mastic, he studied the distribution of nearly identical colloidal granules, with diameters ranging between 0.1 μm and 1 μm, in liquid suspension. By counting the number of granules at various vertical levels using a high-power microscope, he observed that their concentration decreased exponentially with height—analogous to how atmospheric pressure decreases with altitude. The size of the granules was determined both by direct weighing and by applying Stokes' formula.

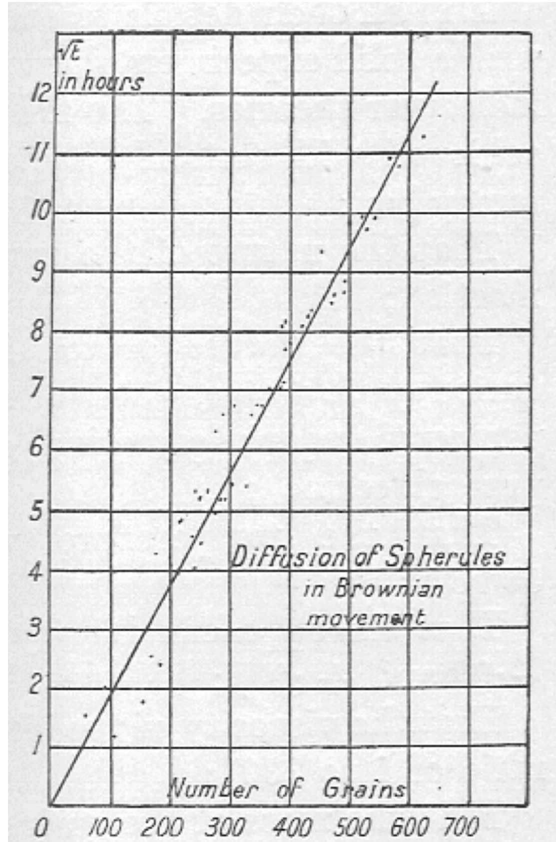

FIG. 3. Perrin's plot showing the linear relation between particle diffusion and $\sqrt{t}$, confirming Brownian motion behavior [13].

Figure 4 reproduces three trajectories of mastic granules, traced by connecting their positions at 30-second intervals. It is the mean square of these segment lengths that confirms Einstein's theoretical formula. One of the trajectories shows 50 consecutive positions of a single granule. While these simplified traces provide a general impression, they greatly underrepresent the highly intricate and tangled nature of the actual particle paths.

From these observations, Perrin demonstrated that the granules behaved statistically like a gas of extremely high molecular weight. This implied that each granule possessed the same average kinetic energy as a molecule of gas or liquid at the same temperature. Such a conclusion provided compelling support for the law of equipartition of energy, suggesting that it holds not only at the molecular level but also for larger particles composed of millions of molecules.

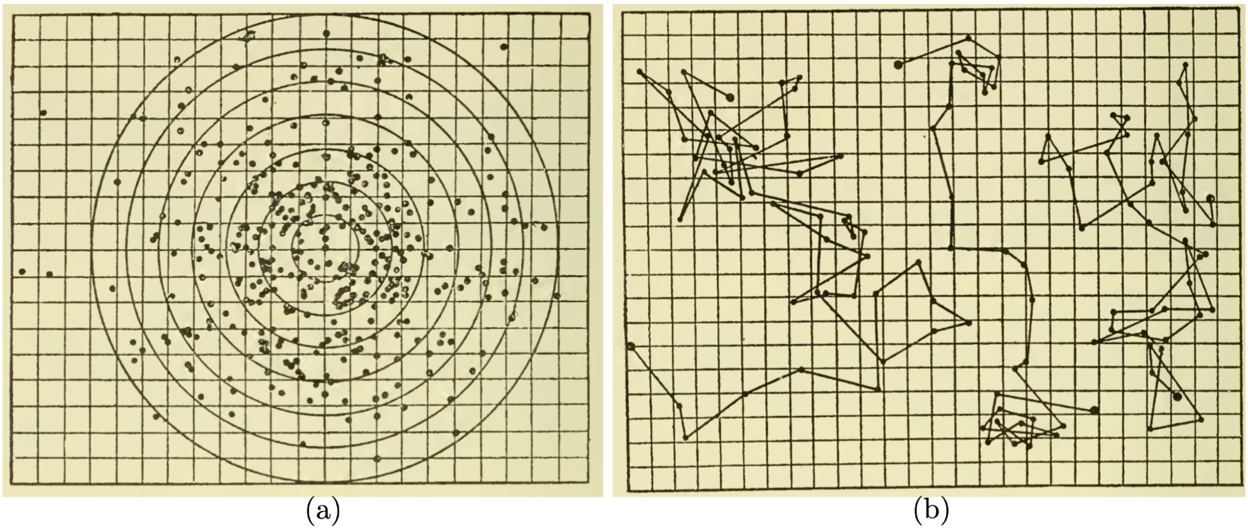

FIG. 4. A two-dimensional random walk tracing the path of a Brownian particle at 30-second intervals. This figure is adapted from Perrin's original observations, where he recorded the positions of colloidal particles over time to illustrate the erratic nature of Brownian motion [7].

Applying this method, he determined Avogadro's number through a series of careful experiments. The table in Fig. 5 from Perrin's 1916 book *Atom* [13] serves to summarize the various phenomena through which N can be calculated, forming a robust body of evidence for the molecular nature of matter. Similar results were obtained in his 1908 experiments, though the numerical values in [7] and [13] differ slightly. Assuming the later version to be more refined, it is included in this report. His work was highly regarded by both Einstein and Smoluchowski [15]. Determining this constant enabled calculations of molecular mass and the fundamental unit of electric charge carried by the hydrogen ion.

Perrin's experiments were both ingenious and meticulous, and they highlighted the subtle behaviors of colloidal granules. By observing large granules of approximately 13 μm in diameter—some of which contained inclusions visible under the microscope—he was able to demonstrate that, in addition to translational Brownian motion, the particles also exhibited spontaneous irregular rotation. Furthermore, the average energy of this rotation was found to be comparable to the energy of translation, thus offering yet another confirmation of the equipartition theorem. Perrin's results were decisive in establishing the kinetic origin of Brownian motion, and numerous later experiments, enhanced by improved techniques, confirmed and refined his findings.

Phenomena observed.		$\frac{N}{10^{22}}$.
Viscosity of gases (van der Waal's equation)		62
Brownian movement	Distribution of grains	68·3
	Displacements	68·8
	Rotations	65
	Diffusion	69
Irregular molecular distribution	Critical opalescence	75
	The blue of the sky	60 (?)
Black body spectrum		64
Charged spheres (in a gas)		68
Radioactivity	Charges produced	62·5
	Helium engendered	64
	Radium lost	71
	Energy radiated	60

FIG. 5. Perrin's experimentally derived values of Avagadro's number N [13] closely matched other theoretical and observational approaches of the time, providing strong support for the atomic theory of matter [7, 13, 14].

4. CONCLUSION

Brownian motion, first observed by Robert Brown as the erratic jitter of pollen grains and later unraveled by the theoretical work of Einstein, Langevin, Smoluchowski, and others, is a pillar of modern physics. Jean Perrin's precise experiments solidified these ideas, confirming the atomic hypothesis and tying microscopic fluctuations to measurable quantities like Avogadro's number. The random walk and dynamical models in this report reveal how thermal energy and viscosity govern this motion, encapsulated in the Stokes-Einstein-Sutherland relation, $D = k_B T/6\pi\eta a$. This equation shows how temperature, fluid viscosity, and particle size dictate the diffusive process at the heart of Brownian motion.

Its applications span far and wide. In physics, it explains colloid behavior and polymer dynamics; in engineering, it shapes microfluidics and nanotechnology. In biophysics, however, Brownian motion is especially vital. It drives the diffusion of proteins within cells, enabling processes like signal transduction and nutrient uptake. For instance, thermal collisions propel enzymes toward substrates, a key step in metabolism. Similarly, bacteria use Brownian motion alongside chemotaxis to navigate toward food sources. Techniques like single-molecule tracking further harness this randomness to study protein interactions, offering insights into cellular function. Far from a mere curiosity, Brownian motion is the subtle force linking the chaotic jostling of particles to the intricate workings of life.

ACKNOWLEDGMENTS

I am struck by the elegance of Einstein's rigorous theory and Langevin's simplified yet profound approach. Equally, I admire Perrin's ingenious and meticulous experiments. Though I initially aimed to explore applications in this project, the brilliance of their seminal work drew me in, shaping this review around its theoretical core. I am deeply grateful to Dr. Igor for granting me this chance to delve into such remarkable science.

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